

PHISHING EMAIL DETECTION AND CYBER THREAT INTELLIGENCE SYSTEM

A Literature Survey and Comparative Review

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Abstract—Phishing email remains one of the most common entry points for data breaches, ransomware, and financial fraud, while the volume and sophistication of cyber threats increasingly demand structured, shareable intelligence rather than isolated point defenses. Rapidly evolving social-engineering tactics, adversarial evasion of automated filters, and fragmented threat data across organizations motivate the design of an integrated system that both detects phishing emails accurately and contributes to a broader cyber threat intelligence (CTI) ecosystem. This paper presents a literature survey of fifty research works spanning rule-based and heuristic phishing filters, machine-learning and deep-learning email classifiers, transformer and large-language-model-based detectors, malicious URL and phishing-website analysis, business email compromise (BEC) and social-engineering detection, malware and attachment sandboxing, threat-intelligence sharing standards such as STIX/TAXII and MISP, dark-web and OSINT-based intelligence collection, security-awareness gamification, and cloud-native SOAR/XDR security-operations architectures. The objectives, methodologies, datasets, and key findings of the surveyed works are summarized and compared using a structured comparison table. Based on this review, the paper identifies the limitations of existing systems and the research gaps that motivate the design of an integrated Phishing Email Detection and Cyber Threat Intelligence System, and outlines future research directions for such a system.

Keywords—Phishing detection, cyber threat intelligence, machine learning, deep learning, natural language processing, large language models, business email compromise, STIX/TAXII, dark web, malware sandboxing, security orchestration, literature survey

I. INTRODUCTION

Phishing email remains one of the most persistent and cost-effective attack vectors in the cyber-threat landscape, serving as the initial foothold for credential theft, ransomware deployment, and large-scale data breaches. Despite decades of research and widespread deployment of spam filters, attackers continue to adapt — crafting messages that exploit trust, urgency, and authority, and increasingly leveraging generative AI to produce grammatically flawless, highly personalized lures that evade both human suspicion and automated detection. At the same time, individual detection systems rarely operate in isolation from the broader threat landscape: an indicator observed at one organization is often relevant to many others, motivating the parallel discipline of cyber threat intelligence (CTI), which structures, shares, and operationalizes knowledge about active threats across organizational boundaries.

A Phishing Email Detection and Cyber Threat Intelligence System aims to close the gap between isolated, single-signal phishing filters and the broader ecosystem of shared threat knowledge by combining accurate, explainable email- and URL-level detection with structured ingestion of external threat intelligence. Before designing such a system, it is essential to understand what has already been attempted in the literature, what technologies and datasets have been used, what results have been achieved, and what limitations remain. This paper therefore conducts a structured literature survey of fifty research works

relevant to the assigned problem statement, drawn from IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, arXiv, Nature, and other scholarly repositories.

The remainder of this paper is organized as follows. Section II describes the review methodology. Section III presents the literature survey, grouped into eleven thematic sub-areas. Section IV presents a comparative analysis of representative papers. Section V discusses the limitations of existing approaches. Section VI identifies the research gaps. Section VII outlines proposed future directions for the Phishing Email Detection and Cyber Threat Intelligence System, and Section VIII concludes the paper.

II. REVIEW METHODOLOGY

Papers were identified using keyword searches such as "phishing email detection machine learning," "deep learning phishing classification," "phishing URL detection," "business email compromise detection," "large language model phishing," "cyber threat intelligence STIX TAXII," "dark web OSINT threat intelligence," "malware sandbox detection," "security awareness gamification phishing," and "SOAR XDR security operations." Papers were selected based on relevance to phishing detection or cyber threat intelligence, recency, and the presence of a clearly stated objective, methodology, and outcome. A total of fifty papers, spanning peer-reviewed journals, IEEE/ACM conference proceedings, and reputable preprint archives, were shortlisted and reviewed in depth; their objectives,

methods, datasets/tools, and findings were extracted and compared, as reported in Sections III and IV.

III. LITERATURE SURVEY

A. Rule-Based and Heuristic Phishing Detection

Early anti-phishing systems relied on blacklisting, whitelisting, and hand-crafted heuristics rather than learned models. PhishNet demonstrated predictive blacklisting that expands a known list of malicious URLs by exploiting common registration and hosting patterns among phishing sites [1]. Content-based heuristics such as CANTINA and PhishShield examined lexical and page-structure cues — brand-keyword frequency, age of domain, and suspicious form fields — to flag fraudulent pages without requiring labelled training data [3], [4]. Lightweight search-based detectors similarly used a handful of low-cost URL and search-engine-ranking features to achieve fast, resource-efficient screening suitable for desktop or gateway deployment [3]. A broader survey of phishing-email detection techniques catalogues these heuristic approaches and observes that, while inexpensive and interpretable, they degrade quickly against zero-day and highly obfuscated campaigns, motivating the shift toward learned models described in the following subsections [4].

B. Machine Learning for Phishing Email Detection

A substantial body of work applies classical supervised learning to labelled corpora of phishing and legitimate emails. Comparative studies combining one-hot encoding with Naive Bayes, Support Vector Machines, and ensemble classifiers report that feature-engineered machine-learning pipelines can reliably separate phishing from legitimate mail on benchmark corpora such as Enron and Nazario collections [5]. Hybrid architectures that couple Support Vector Machines with NLP-derived linguistic features and probabilistic neural networks improve robustness to paraphrased phishing text [7]. Two systematic literature reviews specifically focused on NLP-driven phishing-email detection synthesize dozens of such studies and find that TF-IDF and n-gram feature extraction, combined with classical classifiers, remain competitive baselines even as deep-learning methods have emerged, though they generalize poorly to previously unseen social-engineering templates [8], [9]. Empirical comparisons of Random Forest, Naive Bayes, and Decision-Tree classifiers on phishing corpora further show that ensemble methods consistently outperform single classifiers on precision and recall [6].

C. Deep Learning and Transformer-Based Phishing Email Detection

Deep-learning approaches move beyond hand-crafted features toward end-to-end representation learning. Two independent systematic reviews of deep-

learning-based phishing-email detection catalogue convolutional, recurrent, and hybrid CNN-RNN architectures, finding that models combining spatial and sequential feature extraction consistently exceed 95% accuracy on standard datasets while raising concerns about dataset overlap and limited real-world generalization [10], [11]. A comparative study of 1D-CNN, LSTM, Bi-LSTM, GRU, and Bi-GRU variants confirms that recurrent augmentation of convolutional base models yields incremental but consistent accuracy gains [12]. HELPHED, a hybrid ensemble learning framework, combines multiple deep architectures to reduce the false-negative rate on sophisticated spear-phishing samples [13]. More recent work fine-tunes transformer language models: an improved phishing/spam/ham detector (IPSDM) built on fine-tuned BERT variants and a related BERT-classification study both report accuracy, precision, and recall above 0.985 on augmented, paraphrase-based test sets, indicating that contextual language models better capture the persuasive and rhetorical cues that heuristic and classical ML methods miss [14], [15].

D. Malicious URL and Phishing Website Detection

Because phishing emails typically deliver a malicious link, a parallel research stream targets URL and web-page classification. Sahingoz et al. demonstrated that lexical, host-based, and word-vector features extracted purely from the URL string, without visiting the destination page, can achieve high accuracy using Random Forest and other classifiers, enabling real-time and low-latency screening [16]. Character-level and word-level deep architectures such as URLNet learn URL representations directly from raw characters using convolutional networks, avoiding manual feature engineering entirely [17]. Transformer-based URLTran adapts BERT-style attention to the URL-classification task and reports improved detection of previously unseen phishing domains compared to LSTM baselines [18]. A character-level CNN model for phishing-URL classification further shows that treating the URL as a character sequence, rather than a bag of hand-crafted features, is robust to typosquatting and homoglyph obfuscation [19]. For deployment on resource-constrained clients, PhishLang uses a distilled MobileBERT variant to analyze page source code and URL features entirely on-device, achieving real-time, explainable phishing-website detection without server-side inference [20].

E. Multimodal and Large-Language-Model-Based Phishing Detection

The emergence of large language models (LLMs) has opened a new line of phishing-detection research that reasons over both text and visual page structure. Multi-modal LLM frameworks combine screenshot analysis with HTML and URL text to identify brand impersonation and visually deceptive phishing pages

that evade purely lexical detectors [21]. Comparative studies of prompt engineering versus fine-tuning report that task-specific fine-tuning of LLMs generally outperforms zero-shot or few-shot prompting for phishing classification, though prompting remains attractive where labelled data is scarce [22]. MultiPhishGuard proposes a multi-agent LLM architecture in which specialized agents independently analyze sender reputation, linguistic persuasion cues, and embedded links before a coordinating agent renders an explainable verdict, improving both accuracy and analyst trust relative to single-model classifiers [23]. Complementary work applying BERT-based transformers to phishing-email classification and evaluating stylometric detectors against AI-generated phishing text confirms that as generative models are increasingly used by attackers to author phishing content, detection systems must themselves be evaluated against LLM-generated, rather than only historical, attack samples [24], [25].

F. Business Email Compromise and Social-Engineering Detection

Business Email Compromise (BEC) attacks rely on impersonation and persuasive language rather than malicious links or attachments, making them largely invisible to conventional URL- or payload-based filters. Cidon et al. describe a high-precision BEC detector that models sender behavior, communication graphs, and linguistic anomalies to flag executive-impersonation attempts with low false-positive rates [26]. Writing-style analysis has been proposed as a complementary signal, comparing an incoming email's stylometric fingerprint against a sender's historical writing patterns to detect account takeover or spoofing [27]. A transformer-based BEC detector combining BERT embeddings with a BiLSTM sequence model captures linguistic persuasion cues without relying on explicit technical indicators of compromise [28]. A recent systematic review of BEC detection and prevention research reports that financial losses attributable to BEC have grown into the billions of dollars annually and that most existing tools remain narrowly focused on generic phishing rather than the nuanced impersonation and social-engineering tactics distinctive to BEC [29]. A modular, adaptive BEC-detection system further argues that because BEC lacks stable technical signatures, detection architectures must be continuously retrainable against evolving pretexts, including those authored with generative AI [30].

G. Malware and Attachment-Based Threat Detection

Beyond text and links, a meaningful share of phishing campaigns deliver malicious attachments. Sandboxing remains the dominant technique: suspicious attachments are detonated in an isolated virtual environment so that behavioral indicators, rather than static signatures, determine maliciousness, allowing detection of zero-day and fileless payloads that

signature-based antivirus tools miss [34]. Because full detonation is time-consuming, several works integrate machine learning with sandboxing to accelerate verdicts; a Cuckoo-Sandbox-based study combining CNN and RNN classifiers over API call sequences reports accuracy approaching 99% while substantially reducing analysis latency compared to pure behavioral sandboxing [32]. Complementary research addresses steganographic malware, in which malicious payloads are hidden inside seemingly benign image attachments, surveying detection techniques based on statistical and deep-learning image analysis [31]. Patent literature on machine-learning-based malicious-attachment detectors similarly emphasizes that rule-based signature matching cannot keep pace with continuously obfuscated malware, motivating classifiers trained on structural and behavioral document features rather than static hashes [33].

H. Cyber Threat Intelligence Platforms and Sharing Standards

Detecting individual phishing emails is most effective when paired with structured cyber threat intelligence (CTI) that lets organizations share indicators of compromise (IoCs) and tactics, techniques, and procedures (TTPs) collectively. The OASIS Structured Threat Information eXpression (STIX) and Trusted Automated Exchange of Intelligence Information (TAXII) standards provide a common, machine-readable language and transport protocol for CTI exchange, and the open-source Malware Information Sharing Platform (MISP) both consumes and produces STIX content while exposing it through TAXII servers, enabling near-real-time dissemination of new indicators across a sharing community [35], [36]. An empirical study analyzing roughly six million STIX objects collected over nine years found that industry-wide STIX/TAXII adoption remains fragmented, with a large proportion of shared objects duplicated across sources and daily unique-object volume insufficient to keep pace with the scale of emerging threats [37]. Complementary work extends STIX 2.1 with the CACAO playbook format to share machine-readable, automatable courses of action alongside descriptive threat intelligence, and broader surveys of CTI research note that most published work concentrates on tactical-level indicator sharing while operational- and strategic-level intelligence remain comparatively under-studied [38], [39].

I. Dark Web and OSINT-Based Threat Intelligence

Because phishing kits, stolen credentials, and attack tooling are frequently traded on dark-web forums and marketplaces before they surface in mainstream telemetry, several works treat the dark web as an open-source intelligence (OSINT) source that complements conventional CTI. Kadoguchi et al. apply machine-learning text classification to dark-web forum posts to

automatically retrieve threat-relevant discussions, though the study notes that traditional ML pipelines applied to darknet text achieve comparatively modest accuracy on this noisy domain [40]. DarkBERT, a language model pretrained specifically on dark-web text, is shown to outperform general-purpose language models on threat-relevant classification and information-extraction tasks within this domain [41]. Because text-based CAPTCHA is the most common barrier to automated dark-web crawling, generative-adversarial approaches have been proposed to defeat CAPTCHA challenges at scale for proactive CTI collection [42]. Complementary social-media-based crowdsourcing work shows that early discussion of vulnerabilities and exploits on platforms such as forums and microblogs can be mined to anticipate attacks before they are formally disclosed, while foundational OSINT methodology work catalogues the tools, ethical considerations, and collection workflows appropriate for dark-web-sourced intelligence [43], [44].

J. Security Awareness Training and Gamification

Because a meaningful share of phishing incidents succeed through human rather than technical failure, gamified security-awareness training is a relevant complementary layer for any detection-focused system. A large-scale organizational study finds that gamified e-training measurably improves employees' self-efficacy and their observed response behavior during live phishing simulations, compared with conventional slide-based training [45]. A structured review of security-awareness-training literature from 2021–2024 similarly concludes that adaptive, simulation-based, and gamified training programs reduce phishing susceptibility more effectively than static, one-time awareness sessions, though training fatigue and evolving attacker tactics remain open challenges [46]. Game-based interventions such as Anti-Phishing Phil and related mobile anti-phishing games have been shown to build users' self-efficacy and avoidance

behavior toward phishing links through interactive, low-stakes practice [47]. A broader survey of gamified security-awareness approaches across sectors, including smart-grid and enterprise settings, finds consistent short-term gains in phishing-recognition accuracy, while cautioning that few studies measure whether gains persist over extended periods [48].

K. Cloud-Native SOC Automation: SOAR and XDR Architectures

Finally, several works address the backend architecture required to operationalize phishing and threat-intelligence detections at organizational scale. Security Orchestration, Automation, and Response (SOAR) platforms centralize alert triage, automate routine investigation steps, and orchestrate response workflows across email-security, endpoint, and identity tools, reducing analyst alert fatigue in security operations centers (SOCs) [49]. Extended Detection and Response (XDR) platforms extend this model by unifying telemetry collection and correlation across endpoint, network, identity, and cloud domains within a single detection and response layer, and industry analyses increasingly frame XDR as absorbing much of the correlation functionality previously split between SIEM and SOAR tools [49]. A recent framework for human–AI collaboration in SOC describes how playbook-driven SOAR automation, combined with AI-assisted triage, allows human analysts to focus attention on the highest-confidence and highest-impact alerts while routine phishing and malware events are handled through automated containment workflows [50]. This architectural pattern — independently scalable detection, orchestration, and intelligence-sharing services behind a common API layer — is directly applicable to a phishing-and-CTI platform, where email classification, URL analysis, BEC detection, and threat-intelligence ingestion can each be deployed as separately scalable microservices.

TABLE I

Comparative Analysis of Representative Reviewed Papers

Ref.	Focus Area	Methodology / Technology	Dataset / Tools	Key Findings	Limitations
[1] PhishNet (INFOCOM, 2010)	URL blacklist expansion	Predictive blacklisting via URL-pattern propagation	Known phishing blacklists	Expands limited blacklists to catch related unseen phishing URLs	Cannot generalize beyond structurally similar URLs
[9] Salloum et al. (IEEE Access, 2022)	NLP-based phishing-email detection	Systematic literature review, TF-IDF / classical ML	Multiple email corpora (review)	Synthesizes NLP feature-extraction and classifier trends across studies	Heterogeneous datasets hinder cross-study comparison
[11] Thakur et al. (Electronics, 2023)	Deep-learning phishing-email detection	Systematic review of CNN/RNN/hybrid models	Public phishing-email corpora (review)	DL models consistently exceed 90–95% accuracy on benchmark sets	Risk of dataset overlap; limited real-world validation
[13] Bountakas & Xenakis (J. Netw. Comput. Appl., 2023)	Hybrid ensemble phishing detection	HELPHED: hybrid ensemble deep-learning framework	Public phishing/legitimate email sets	Reduces false negatives on spear-phishing samples vs. single models	Higher computational cost of ensembling

Ref.	Focus Area	Methodology / Technology	Dataset / Tools	Key Findings	Limitations
[15] Fang et al. (IEEE Access, 2019)	Phishing-email text classification	Improved RCNN with multilevel vectors + attention	Nazario / public phishing corpus	Attention mechanism improves detection of subtle phishing cues	Attention weights not evaluated for interpretability by users
[16] Sahingoz et al. (Expert Syst. Appl., 2019)	URL-based phishing detection	NLP + Random Forest on URL lexical/host features	Custom large-scale URL dataset	High accuracy without needing to visit the destination page	Feature set may be evaded by adversarial URL crafting
[18] Maneriker et al. URLTran (2021)	Transformer-based URL detection	BERT-style transformer over URL tokens	Enterprise phishing-URL logs	Detects previously unseen phishing domains better than LSTM baselines	Requires substantial labelled training data
[20] Roy & Nilizadeh PhishLang (arXiv, 2024)	Client-side phishing-website detection	MobileBERT lightweight ensemble	Live phishing/benign webpages	Real-time, explainable, fully on-device detection	Robustness against adversarial page manipulation still limited
[23] MultiPhishGuard (arXiv, 2025)	LLM-based phishing-email detection	Multi-agent large language model architecture	Public phishing-email corpora	Specialized agents improve accuracy and explainability of verdicts	Inference cost and latency of multi-agent LLM pipeline
[26] Cidon et al. (USENIX Security, 2019)	Business email compromise detection	Behavioral and communication-graph modelling	Enterprise email-account data	High-precision BEC detection with low false-positive rate	Requires historical communication baseline per sender
[29] BEC Systematic Review (ScienceDirect, 2025)	BEC understanding and detection	Systematic literature review	N/A (review)	Catalogues rule-based, ML, and NLP BEC-detection approaches	BEC remains comparatively under-researched vs. generic phishing
[32] Alshmami & Alliheedi (arXiv, 2023)	Malware/attachment detection	CNN & RNN over Cuckoo Sandbox API-call sequences	Malware API-call dataset	Up to ~99% classification accuracy; faster than pure sandboxing	Sandbox evasion by time-aware or environment-aware malware
[36] MISP / STIX-TAXII (misp-project.org)	CTI sharing platform	Open-source CTI platform with STIX export / TAXII sync	Community-contributed IoC feeds	Near-real-time, machine-readable indicator dissemination	Data quality depends on contributing-community diligence
[37] STIX Sharing Study (NDSS, 2024)	CTI-sharing effectiveness	Empirical analysis of ~6M STIX objects (9 years)	Public/commercial STIX feeds	Reveals substantial duplication and fragmented adoption	Daily unique-indicator volume trails threat growth
[41] DarkBERT (ACL, 2023)	Dark-web threat intelligence	Language model pretrained on dark-web text	Dark-web forum corpus	Outperforms general LMs on dark-web classification tasks	Domain-specific pretraining data is difficult to collect and curate
[45] Gamified Training Study (ScienceDirect, 2024)	Security-awareness gamification	Field study of gamified e-training	Enterprise employee cohort	Improves self-efficacy and real phishing-simulation response	Long-term retention beyond study period not measured
[49] SOAR/XDR Overview (IBM, 2025)	SOC automation architecture	Security orchestration, automation, and response	Industry SOC deployments (report)	Reduces alert fatigue; centralizes cross-tool response workflows	Integration complexity across heterogeneous security tools

V. LIMITATIONS OF EXISTING APPROACHES

Across the reviewed literature, several recurring limitations emerge. First, most phishing-detection systems remain single-vector: email-text classifiers do not analyze embedded URLs or attachments in depth, and URL- or website-focused detectors rarely model the linguistic persuasion cues that drive business email compromise, forcing organizations to stitch together multiple disconnected tools. Second, threat-intelligence-sharing platforms such as STIX/TAXII and MISP are largely designed for analyst-to-analyst exchange rather than for direct integration into end-user-facing detection pipelines, so the trust and coverage benefits of CTI sharing are not always experienced at the point of email delivery. Third, many machine-learning and deep-learning phishing detectors are evaluated on public

benchmark corpora that are aging, imbalanced, or overlap across studies, raising concerns about generalization to novel, LLM-generated phishing campaigns. Fourth, dark-web and OSINT-based intelligence collection remains comparatively immature relative to clear-web CTI, with only a small fraction of surveyed systems substantively engaging this source despite its value for early-warning detection. Finally, explainability, adversarial robustness, and the operational cost of multi-agent or multi-model pipelines are consistently flagged as open concerns rather than solved problems across the surveyed works [10], [22], [37], [46].

VI. RESEARCH GAPS

- Lack of a single, integrated platform that combines email-text classification,

URL/website analysis, BEC-specific behavioral detection, and CTI ingestion within one coherent system.

- Limited real-time integration between phishing-detection engines and structured threat-intelligence feeds (STIX/TAXII/MISP), so newly observed indicators are not immediately reflected in email-filtering decisions.
- Insufficient benchmarking of detection models against LLM-generated and adversarially paraphrased phishing content, which increasingly differs from historical training corpora.
- Sparse evaluation of dark-web and OSINT-derived intelligence as an early-warning input to phishing-detection pipelines, despite its demonstrated value for anticipating emerging campaigns.
- Underexplored integration of gamified security-awareness modules with automated detection systems, leaving the human and technical layers of defense largely disconnected.
- Limited attention to explainable, resource-efficient architectures that can run multi-signal detection (text, URL, attachment, sender-behavior) within the latency constraints of live email delivery.

VII. PROPOSED DIRECTIONS FOR FUTURE WORK

Building on the identified gaps, a Phishing Email Detection and Cyber Threat Intelligence System should combine (i) a transformer- or LLM-based email-text classifier kept current through continual fine-tuning on recent and adversarially generated phishing samples, in the spirit of the multi-agent and comparative deep-learning approaches surveyed [12], [23]; (ii) a lightweight URL/website-analysis module suitable for real-time, client-side deployment, drawing on MobileBERT-based and character-level techniques [19], [20]; (iii) a behavioral BEC-detection layer that profiles sender writing style and communication graphs to catch attacks lacking malicious links or attachments [26], [28]; (iv) native ingestion of STIX/TAXII and MISP threat-intelligence feeds so that externally observed indicators of compromise are reflected in filtering decisions with minimal delay [35], [36]; (v) an optional dark-web and OSINT monitoring component to surface emerging phishing kits and credential-trading activity before they reach mainstream campaigns [40], [41]; and (vi) a gamified, adaptive security-awareness layer that reinforces detection with user-facing training informed by an organization's own recent near-miss and simulated-phishing data [45], [48]. Architecturally, a

cloud-native microservices design orchestrated through a SOAR/XDR-style automation layer is recommended so that each module — classification, URL analysis, BEC detection, CTI ingestion, and awareness training — can be developed, scaled, and secured independently [49], [50]. Future empirical work should validate such an integrated system on large-scale, organizationally diverse email traffic and should report explainability, latency, and adversarial-robustness metrics alongside raw accuracy.

VIII. CONCLUSION

This paper surveyed fifty research works relevant to the design of a Phishing Email Detection and Cyber Threat Intelligence System, covering rule-based and heuristic filtering, machine-learning and deep-learning email classification, transformer and LLM-based detection, malicious URL and website analysis, business email compromise detection, malware and attachment sandboxing, cyber-threat-intelligence sharing standards, dark-web and OSINT-based intelligence, security-awareness gamification, and cloud-native SOC automation. While substantial progress has been made in each individual sub-area, the literature reveals a clear gap: no reviewed system tightly integrates accurate, explainable phishing and BEC detection with real-time threat-intelligence sharing and proactive dark-web monitoring in a single, operationally deployable platform. Addressing this gap, as outlined in Section VII, represents a concrete and well-motivated direction for the proposed system.

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